



Introduction to Structural Health Monitoring

MENG 4050

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Intro to the Fundamentals of Structural Health Monitoring

- 1. Condition Monitoring Techniques
- 2. Condition-Based Maintenance
- 3. Lead-Time-to-Maintenance (LTM)





Part One

Condition Monitoring

- Basic concepts



Condition Monitoring

Monitoring of machine and process parameters:

the best indicator of machine health condition and its performance e.g., car is the dashboard providing information of engine revolution per minute, coolant level and temperature, parking brake, engine condition, etc.

Ref: Industrial Approaches in VBC Monitoring, Jyoti Sinha

Temperature Monitoring



Temperature Monitoring:

- Used in many cases to identify the defect such as mechanical rubbing in machines.
- Defects in anti-friction bearings, leak in pipe, etc.

A temperature sensor can measure the temperature at a point of machine and structure.

Thermography:

Temperature distribution of a machine or structural surface. Based on the infrared energy emission to measure surface temperature.

- identify the “hot spot” quickly
- Identify the hot spot in the main electrical circuit board (looseness in electric circuits).



Lubricant monitoring

Lubricant monitoring:

Lubricant condition keeps on deteriorating during its use which can impact the machine performance;

1. Lubricant properties

Monitoring concentration of additives and other properties of the lubricant.

2. Debris analysis

Measure number of metal particles and their sizes in the lubricant. The wear particles from the rubbing surface in the lubricant can be used to identify the defects in bearings, gearboxes, etc.

Leak detection monitoring



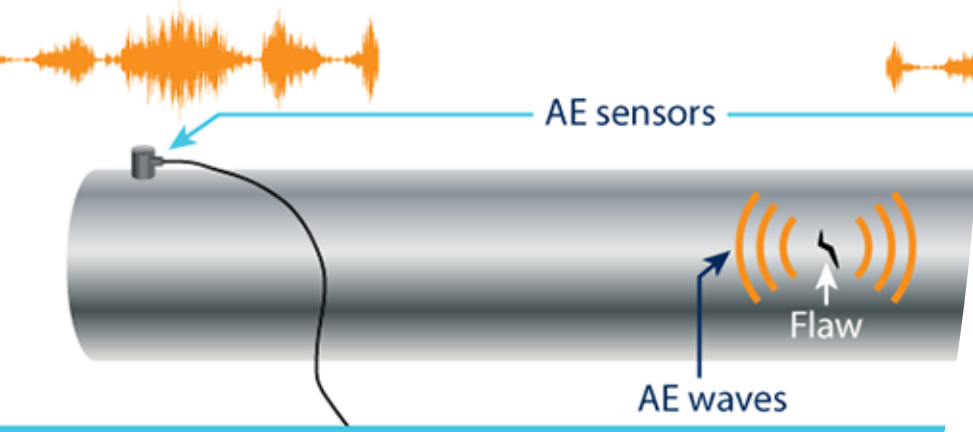
Leak detection monitoring:

The leak detection in the pipe conveying fluid is essential for several applications.

1. Pressure sensor, acoustics wave reflection, etc. can be used to detect the leak.
2. Ultrasound technique is useful tool, but it is a time-consuming process for long pipeline.
3. The tracer gas technique can also be used at certain time interval.

Noise and Acoustic Emission

HOW ACOUSTIC EMISSION TESTING



Noise monitoring:

- Microphones or hydrophones are used for this purpose depending upon the surrounding medium. This is found to be useful for monitoring machines, damage in pipelines, etc.

Acoustic emission monitoring:

It is observed that the occurrence of crack, propagation of damage in any structures, bearing defects, etc. generally release energy at very high frequency range (more than 70–80 kHz). Such signals can be picked up by the acoustic emission sensor;

- The detection process is heavily dependent on the resonance frequency of the sensor.
- appropriate selection of the sensor is essential for any particular application.

Vibration monitoring

Vibration monitoring:

Vibration monitoring is widely used in the CM of machines and structures in industries.

vibration can detect change in any objects (machines or structures) that can be explained through the mechanical vibration theory;

vibration measurements, tests and analysis are tools for design, qualification, design optimization, condition monitoring of structures and machines, vibration insulation and control.





Part Two

Maintenance Strategies

- Basic concepts

Maintenance Strategies

A range of maintenance strategies is available to the maintenance manager. They are:

1. **Breakdown Maintenance (BM)** or Run to failure: Here the maintenance activity is carried out after the failure of items. It is good to select those items of the plant that must have minimal impact on plant safety and production, e.g., electric bulbs.
2. **Planned Preventive Maintenance (PPM)**: This approach is useful for those items that have predictive life; hence these items can be replaced at a fixed time interval related to their life. But this approach may have risks, such as the item may be replaced too soon, item may fail before replacement, etc.
3. **Condition-based Maintenance (CBM)**: This is a much better approach to avoid failure by detecting defect at the early stage through the CM. The predication of the defect or degradation or deterioration by the CM can triggers the maintenance activity to rectify the defect; hence the approach is known as the CBM (or Predictive Maintenance). It has following benefits:
 1. Avoid unplanned shutdown due to breakdown and its consequential damage can be avoided
 2. Reduction in the failure rates and improving plant availability and reliability
 3. Reduction in spares inventory
 4. Reduction in unnecessary work due to sudden breakdown
 5. Possible efficient scheduling of repair/maintenance work to minimize downtime.

Maintenance Strategies

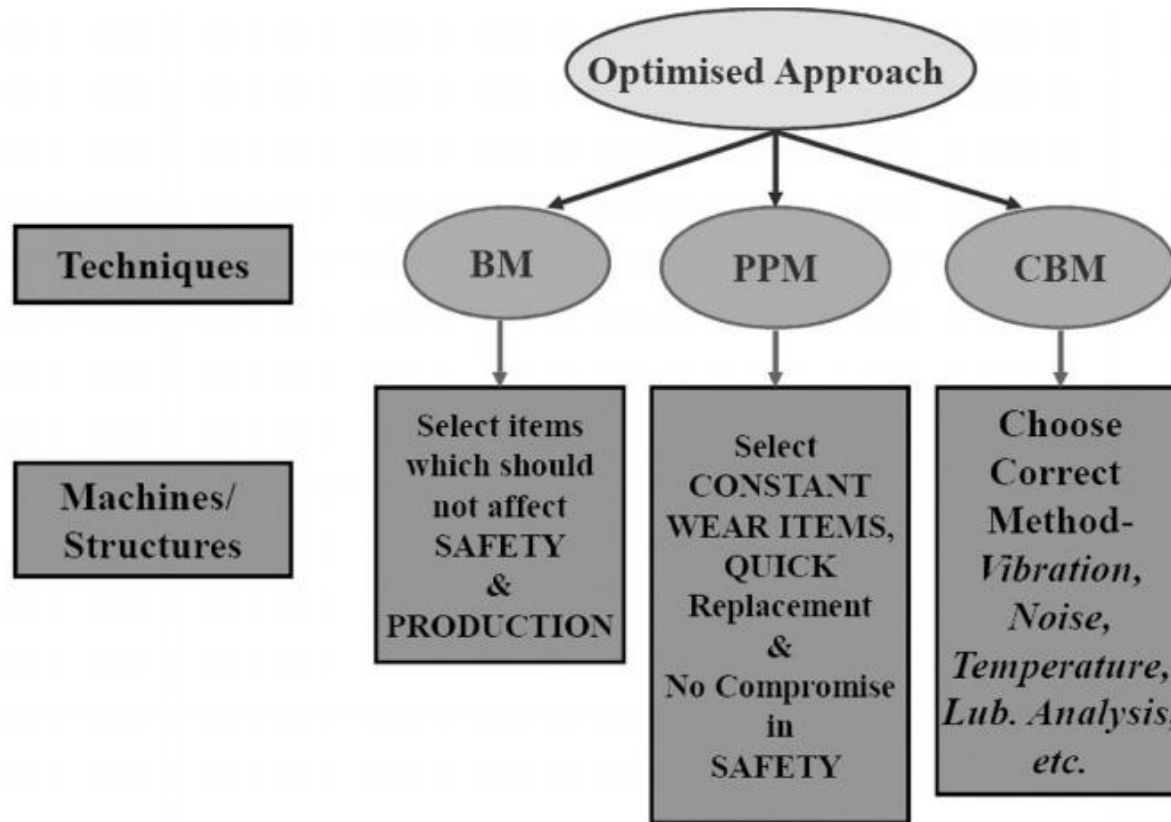


FIGURE 1.4 Viable approaches to minimize maintenance activities and cost.

Condition-based Maintenance

The CM should not be applied to all machines/structures/piping, in a plant because this may not be cost-effective approach; optimized approach is used, as illustrated in Figure 1.4.

The selection of the items (machines or any other components) for the CM and CBM within a plant should follow the steps given in Figure 1.5.

It is important to consider the following three elements for the selection of critical items.

1. Overall impact on the plant safety
2. High capital value, high repair cost in case of any failure
3. Potential production loss, not meeting the required demands

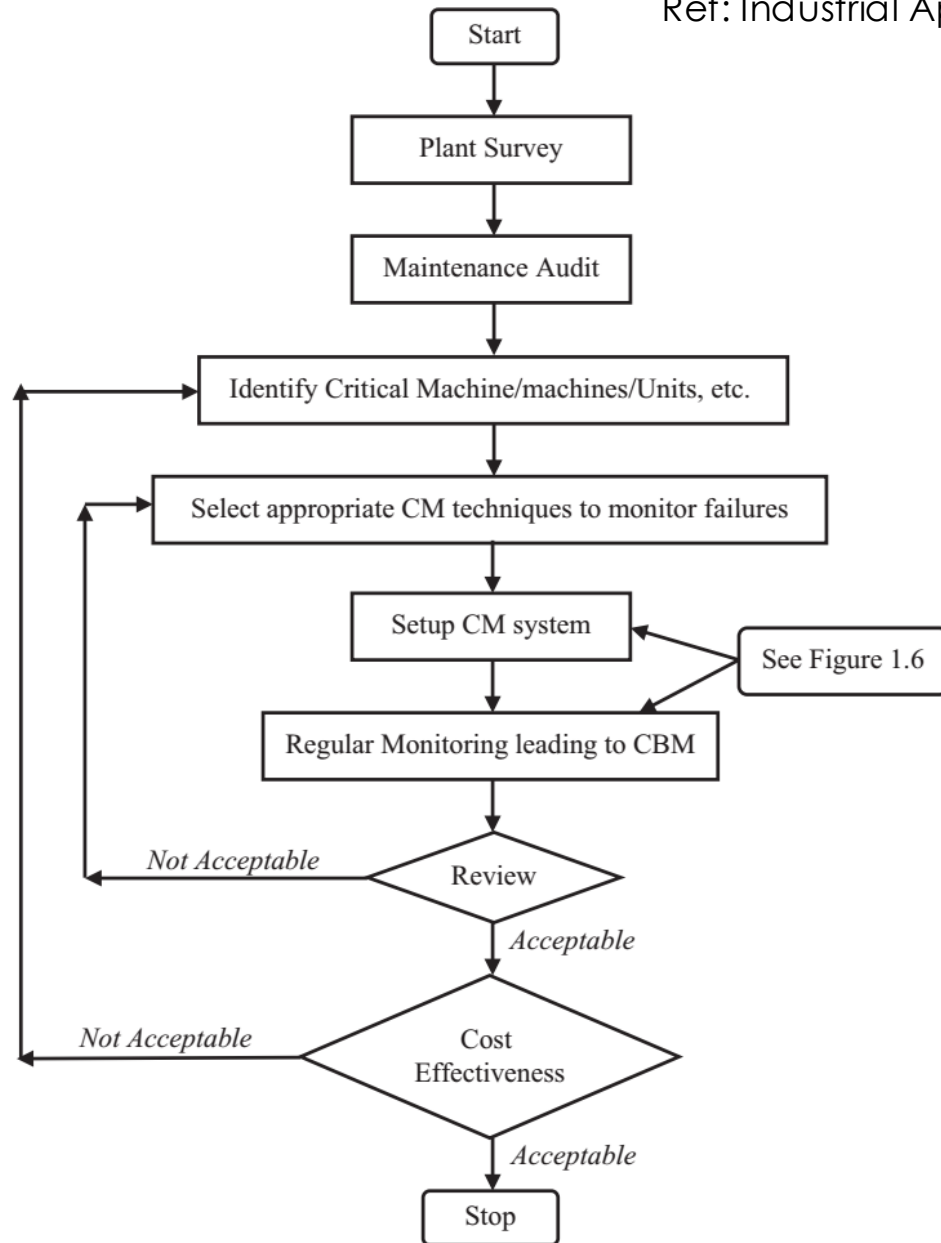


FIGURE 1.5 Steps to select appropriate machine/machines for the CBM.

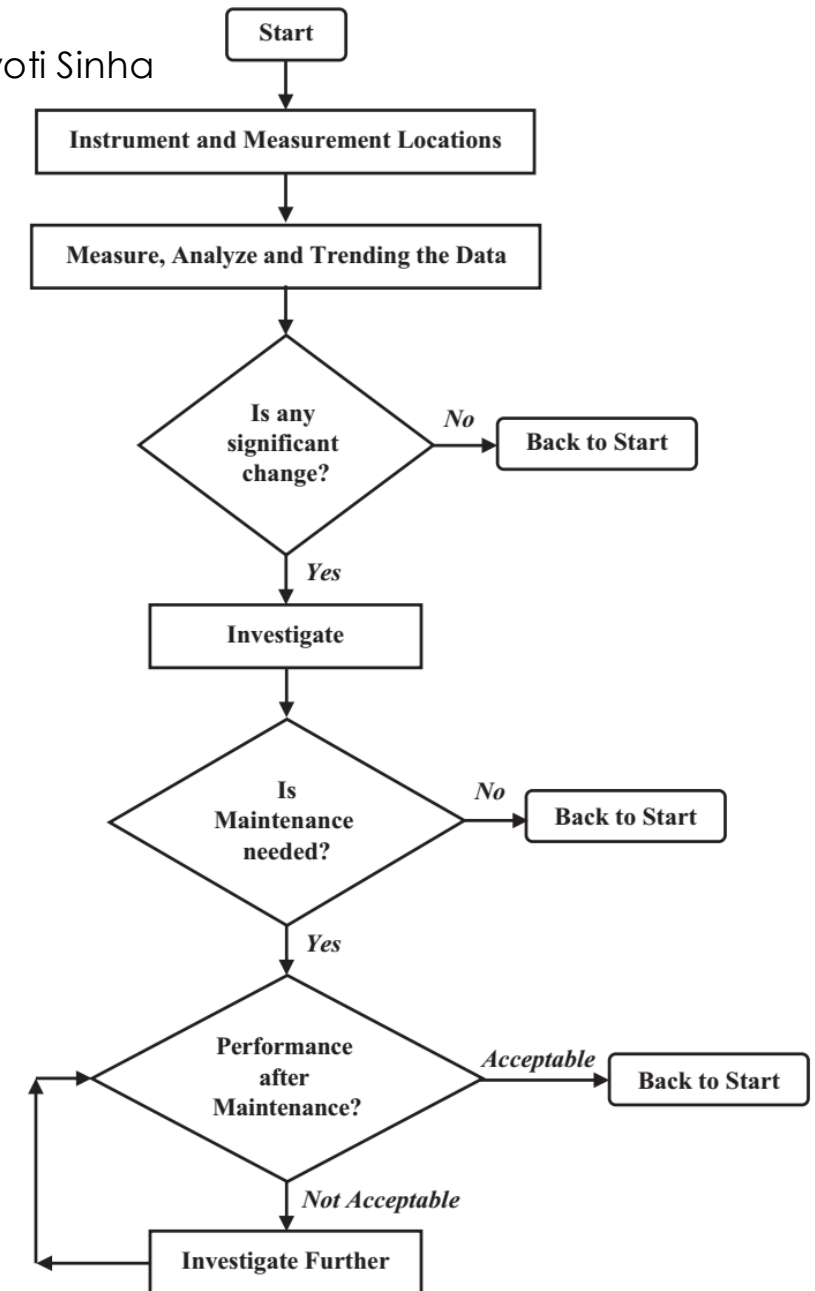


FIGURE 1.6 Simple steps in the CM leading to the CBM.



Part Three

Lead-Time-To-Maintenance

- Basic concepts

Lead-Time-To-Maintenance (LTM)

Lead-Time-To-Maintenance (LTM) After the critical items are selected, the steps given in Figure 1.6 should be followed to monitor the items and perform the maintenance activity if needed.

1. When is it best to carry out the maintenance work to rectify the defect?
2. Should it be immediately after identification of the defect by the CM system?
3. Can the maintenance personnel wait for some time to organize the maintenance activity without affecting the plant safety?

Reliability and Maintenance index (RMI)

Based on machine performance and CM data one should define reliability and maintenance index (RMI):

1. has allowable and alarm limit
2. when this RMI shows increasing trend, extrapolation used to predict the lead time to reach alarm level

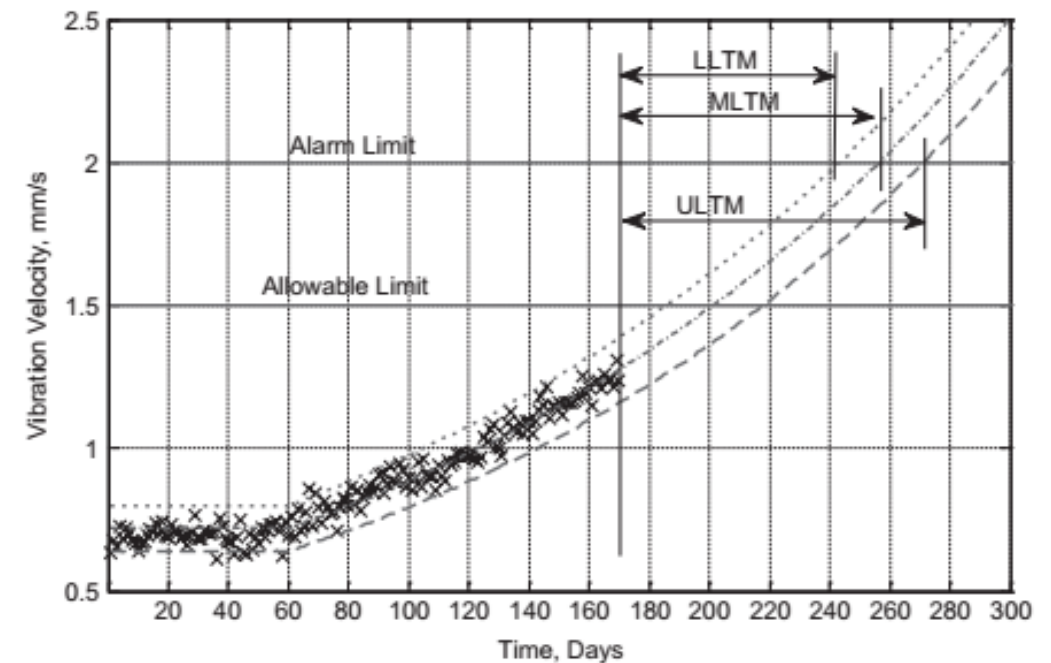


FIGURE 1.8 An example to estimate the LTM.

LEAD TIME TO MAINTENANCE

Two predictions exist:

Lower LTM (LLTM)

This simply means that the earliest maintenance can be planned by the LLTM with minimal risk of failure by this time.

Upper LTM (ULTM)

The ULTM is the possible maximum time that the maintenance personnel can wait for the maintenance activity, but this may have some of risk of failure.

- it is good to have the predicated values of the LTM for all critical items within a plant.
- This help to schedule a best possible period for the plant outage (or shutdown) to carry out all maintenance activities.

Example

the overall machine vibration velocity represents the RMI for the machine considered.

The allowable and alarm limits for the vibration velocity are 1.5 and 2 mm/s respectively for this machine.

The trend of the measured vibration values are shown in Figure 1.8. It is clear from the figure that there is no significant change in the vibration up to 60 days and then shows an increasing trend.

The vibration measurements are carried out up to 170 days.

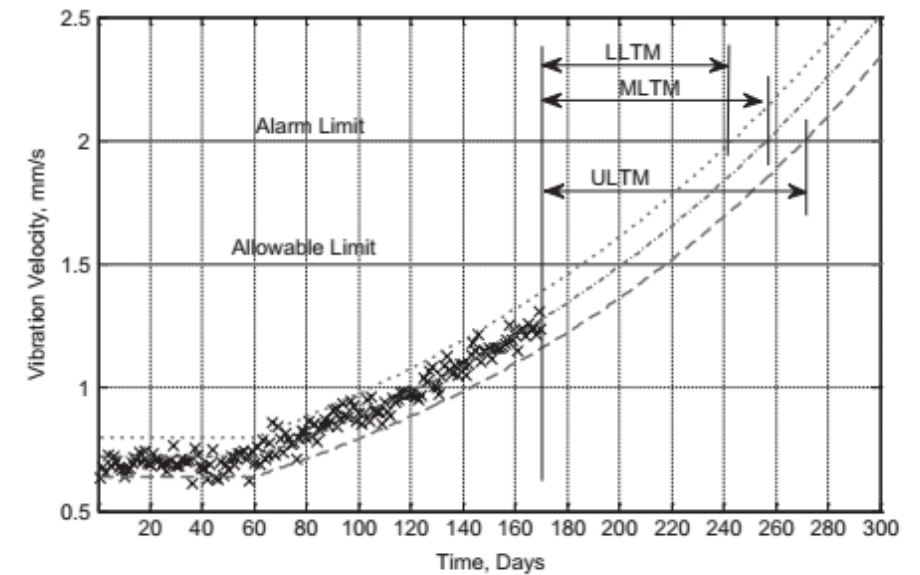


FIGURE 1.8 An example to estimate the LTM.

Example

$$V(t) = V_o \text{ when } 1 \leq t \leq t_o$$

$$V(t) = V_o e^{\lambda(t-t_o)} \text{ when } t \geq t_o$$

TABLE 1.1

List of Parameters Used and Estimated LTM Values

	V_o	t_o	λ	LTM
LLTM	0.80 mm/s	60 days	0.0050	73 days
ULTM	0.64 mm/s	60 days	0.0054	101 days
MLTM	0.072 mm/s	60 days	0.0052	86 days

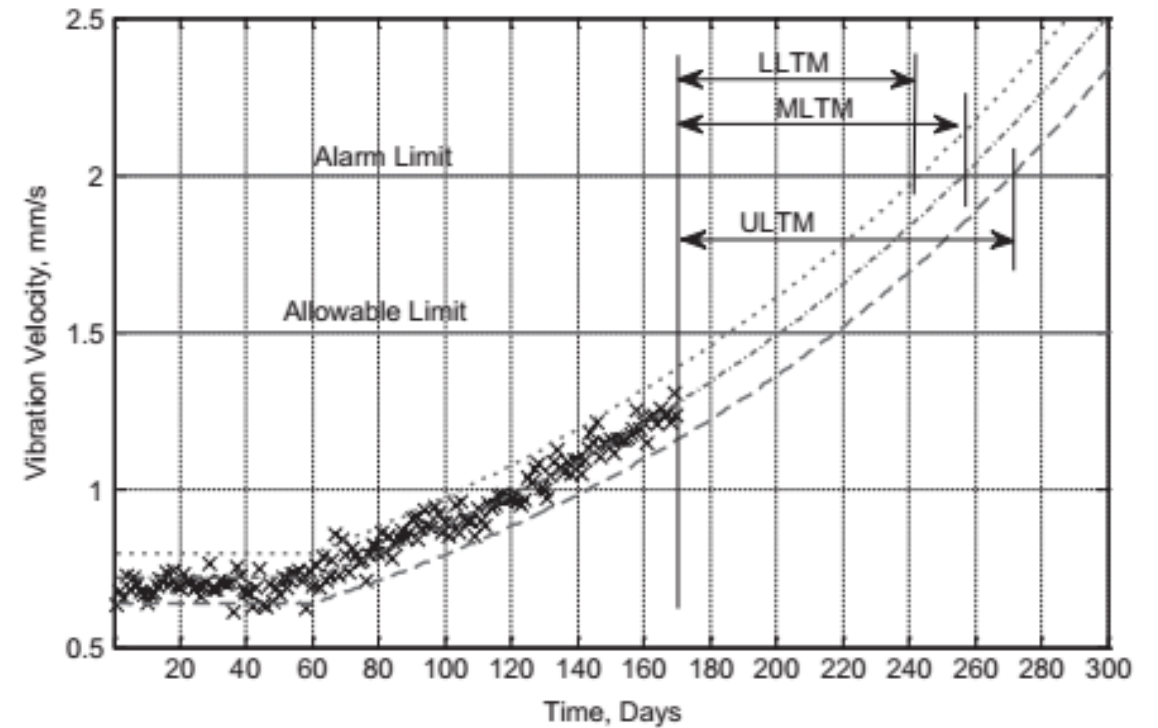


FIGURE 1.8 An example to estimate the LTM.



Questions

Office hours:

Please send all questions through course message